
Discussion 12

***Note:** Your TA will probably not cover all the problems on this worksheet. The discussion worksheets are not designed to be finished within an hour. They are deliberately made slightly longer so they can serve as resources you can use to practice, reinforce, and build upon concepts discussed in lectures, discussions, and homework.*

This Week's Cool AI Demo/Video:

Gemini 3 video chat: https://play.google.com/store/apps/details?id=com.google.android.apps.bard&hl=en_US

Google's Cursor clone just launched: <https://antigravity.google/download>

Sunday came out of stealth: <https://www.youtube.com/watch?v=jj0fpsMRhL4>

1 Supervised Fine-tuning for Vision Language Models

A vision language model (VLM) is a model that integrates visual and textual data, enabling it to understand and process both images and text. By combining a vision encoder and a large language model (LLM), VLMs can perform tasks like generating captions for images, answering questions about an image, and even creating images from text descriptions.

To integrate a **Vision Encoder** that processes an image and outputs patch embeddings of dimension D_{vis} and an **LLM** that expects token embeddings of dimension D_{txt} , we typically define a **Projection Layer** (W_P) that projects visual embeddings into the text embedding space. Supervised Fine-tuning (SFT) is then performed with a small image-text dataset to align the visual features with the language model's embedding space.

- (a) **Projection Dimensions:** Suppose the Vision Encoder outputs an embedding H_{vis} of size $D_{vis} = 1024$ and the LLM expects embeddings of size $D_{txt} = 4096$. What must be the dimensions of the weight matrix, W_P , to project the visual features to the text space if we left-multiply the image embedding by the projection matrix?

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- (b) **Freezing Strategies:** During the initial alignment stage (and often during SFT), it is standard practice to keep the Vision Encoder and the LLM **frozen**, updating only the projection layer. Give two distinct reasons why we freeze the backbone models.
- (c) **Image Token Granularity:** The classify token ([CLS]) is a special token that represents the entire input sequence for a classification task. We can choose to project only the Vision Encoder's classification token into the LLM embedding space, or we can project the entire sequence of patch tokens. What is the primary trade-off for each of these two approaches?
- (d) **The Role of SFT Data:** Explain why we cannot simply rely on the pre-training of the individual components and why the SFT stage (using Image-Text pairs) is necessary even if the embedding dimensions are the same.

2 Self-Supervised Learning

(a) Fill in the blanks in the following table.

Method Name	Input Type	Pretext Task	Generative or Discriminative	Loss Function
Autoencoder	Image	Reconstruct the input image.		
Context Encoder	Masked Image	Predict the missing content in the masked region of the input.		
Image Rotation	Rotated Image	Predict the rotation angle applied to the input (0° , 90° , 180° , or 270°).		
SimCLR (Contrastive Learning)	Two Images	Determine whether images are augmentations of the same image or different images.		

(b) When training **context encoders**, we often use a joint loss function:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{adv}}$$

where:

- $\mathcal{L}_{\text{rec}}$ is the reconstruction loss (e.g., MSE) applied to the masked region
- $\mathcal{L}_{\text{adv}}$ is the adversarial loss from a discriminator
- λ is a weighting factor that balances the two terms

(i) If we trained using **only** $\mathcal{L}_{\text{rec}}$, what artifact might we observe in the generated missing region? Why?

(ii) Why does adding the adversarial loss $\mathcal{L}_{\text{adv}}$ help fix this issue?

